

|                                                                                                                                                                                                                                                                                                                                                                                                                   |                                    |         |         |         |         |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|---------|---------|---------|---------|--|
| Given:<br>Due:                                                                                                                                                                                                                                                                                                                                                                                                    | Further Maths (Core Pure)<br>HW 14 |         |         |         | Name:   |  |
| Chapters 1.4 & 1.5: Complex Numbers (roots of equations)                                                                                                                                                                                                                                                                                                                                                          |                                    |         |         |         |         |  |
| You <b>MUST</b> refer to class notes and complete personal study before attempting this homework. Attempt <b>EVERY</b> part of <b>EVERY</b> question. Check your work for errors or omissions before handing in.                                                                                                                                                                                                  |                                    |         |         |         |         |  |
| 91% (A*)                                                                                                                                                                                                                                                                                                                                                                                                          | 76% (A)                            | 66% (B) | 56% (C) | 50% (D) | 40% (E) |  |
| <p><i>Useful resources:</i> Your notes from recent lessons &amp; your text book</p> <p><u>Although most Complex arithmetic can be done on your calculator, show all steps to reach the answer &amp; use calculator for checking results only.</u></p>                                                                                                                                                             |                                    |         |         |         |         |  |
| <b>Questions - do these on lined paper</b>                                                                                                                                                                                                                                                                                                                                                                        |                                    |         |         |         |         |  |
| <p><b>Q1.</b></p> $f(x) = 9x^3 - 33x^2 - 55x - 25$ <p>Given that <math>x = 5</math> is a solution of the equation <math>f(x) = 0</math>, use an algebraic method to solve <math>f(x) = 0</math> completely.</p> <p style="text-align: right;">(5)</p>                                                                                                                                                             |                                    |         |         |         |         |  |
| <p><b>Q2.</b></p> <p>Given that <math>4</math> and <math>2i - 3</math> are roots of the equation</p> $x^3 + ax^2 + bx - 52 = 0$ <p>where <math>a</math> and <math>b</math> are real constants,</p> <p>(a) write down the third root of the equation,</p> <p style="text-align: right;">(1)</p> <p>(b) find the value of <math>a</math> and the value of <math>b</math>.</p> <p style="text-align: right;">(5)</p> |                                    |         |         |         |         |  |
| <p><b>Q3.</b></p> <p>Given that <math>x = \frac{1}{2}</math> is a root of the equation</p> $2x^3 - 9x^2 + kx - 13 = 0, \quad k \in \mathbb{R}$ <p>find</p> <p>(a) the value of <math>k</math>,</p> <p style="text-align: right;">(3)</p> <p>(a) the other 2 roots of the equation.</p> <p style="text-align: right;">(4)</p>                                                                                      |                                    |         |         |         |         |  |

**Q4.**

$$f(x) = 2x^3 - 6x^2 - 7x - 4$$

(a) Show that  $f(4) = 0$

(1)

(b) Use algebra to solve  $f(x) = 0$  completely.

(4)

**Q5.**

The roots of the equation

$$z^3 - 8z^2 + 22z - 20 = 0$$

are  $z_1, z_2$  and  $z_3$ .

(a) Given that  $z_1 = 3 + i$ , find  $z_2$  and  $z_3$ .

(4)

(b) Show, on a single Argand diagram, the points representing  $z_1, z_2$  and  $z_3$ .

(2)

**Q6.**

Given that 2 and  $5 + 2i$  are roots of the equation

$$x^3 - 12x^2 + cx + d = 0, \quad c, d \in \mathbb{R};$$

(a) write down the other complex root of the equation.

(1)

(b) Find the value of  $c$  and the value of  $d$ .

(5)

(c) Show the three roots of this equation on a single Argand diagram.

(2)

**END OF HOMEWORK**

**Total = 37 marks**

Hand your work in with this sheet. Failure to meet your target grade will result in intervention support.

File this question paper in your file, ready for revision and file inspection.